

Inspire the love of learning

WOS

Q1. Perceptron Update Rule

- Given a single-layer perceptron with $w = [0.5, -0.3]$, bias $b = 0.1$, learning rate $\eta = 0.1$. A training sample $x = [1, 2]$ has true label $y = +1$. Compute one update step.

• Step 1: Forward pass:

- Net input: $z = wx + b = 0.5 \times 1 + 2 \times (-0.3) + 0.1 = 0.0$

- Prediction: $\hat{y} = \text{sign}(z) = \text{sign}(0.0) = +1$

• Step 2: Check misclassification

- $\hat{y} = +1 = y \Rightarrow$ Correctly classified $\rightarrow$ No update needed

Now suppose $x = [1, 2]$, $y = +1$ but $w = [0.5, -0.3]$, $b = -0.2$

- $z = 0.5 - 0.6 - 0.2 = -0.3$

- $\hat{y} = \text{sign}(-0.3) = -1 \neq y = +1 \Rightarrow$ misclassified

• Step 3: Update

- $w_{\text{new}} = w + \eta \cdot y \cdot x = [0.5, -0.3] + 0.1 \times 1 \times [1, 2]$
 $= [0.6, -0.1]$

- $b_{\text{new}} = b + \eta \cdot y = -0.2 + 0.1 \times 1 = -0.1$

- $z_{\text{new}} = 0.6 \times 1 + (-0.1) \times 2 + (-0.1) = 0.6 - 0.2 - 0.1$

$= 0.3 > 0 \Rightarrow \hat{y} = +1$

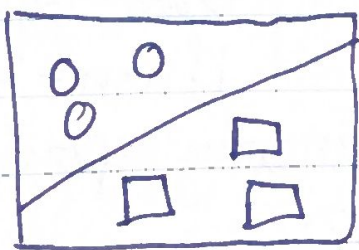
Q2 Linear Separability

What does it mean for a data set to be linear separable?

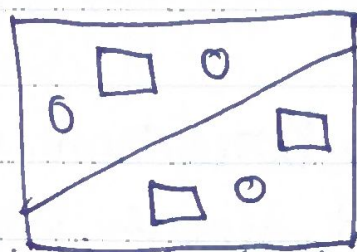
Give example of a linearly separable problem & one that is not.

- A data set is linear separable if there exists a hyperplane that perfectly separate all positive sample from all negative samples with zero training error. Formally:

There exists w, b such that for all $(x_i, y_i): y_i (w \cdot x_i + b) > 0$



Linear Separable



Linear non-separable

Blueangel

Q3. Why XOR fail for single perceptron
 Prove that a single layer perceptron cannot learn XOR. What is the minimum architecture needed?

Proof by contradiction: Assume a perceptron with weights (w_1, w_2) and bias b can solve XOR with threshold 0:

$$(1) \ x = (0, 0) \Rightarrow y = 0 : b \leq 0$$

$$(2) \ x = (0, 1) \Rightarrow y = 1 : w_2 + b > 0$$

$$(3) \ x = (1, 0) \Rightarrow y = 1 : w_1 + b > 0$$

$$(4) \ x = (1, 1) \Rightarrow y = 0 : w_1 + w_2 + b \leq 0$$

$$(2) + (3) : (w_1 + w_2 + b) + b > 0 \Rightarrow w_1 + w_2 + b > -b \geq 0$$

($b \leq 0$ implies $-b \geq 0$)

$\Rightarrow w_1 + w_2 + b > 0$. Which $\nless (4) \Rightarrow$ can't learn XOR

Minimum architecture needed: A 2-layer MLP: 2 inputs
 $\rightarrow$ 2 hidden neuron (non-linear activation)
 $\rightarrow$ 1 output neuron.

Q4 Perceptron Rule vs. Delta (Gradient) Rule

Compare the perceptron rule with Delta (LMS) Rule.

Property	Perceptron Rule	Delta Rule
Output y_{pred}	Discrete: $\text{sign}(w \cdot x + b)$ $\text{in } \{-1, +1\}$	Continuous / Linear: $w \cdot x + b$
Update Condition	Only misclassified samples	Every sample
Update Formula	$\Delta w = \eta \cdot y \cdot x$	$\Delta w = \eta \cdot (y - y_{pred}) \cdot x$
Loss & Convergence	No loss function, con converges only if data is linearly separable	Minimize MSE loss, converges to best least-squared fit even if non-separable

- Prefer Perceptron: when data is linearly separable & need a fast, simple binary classifier.

- Delta Rule: data contains noise or overlap, Blueangel
 or as the mathematical foundation for multi-layer Backpropagation network